Continuous Operation of a Myoelectric Hand Prosthesis using the TD3 Reinforcement Learning Algorithm

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Abstract: Achieving proportional, real time operation of prosthetic hands from raw electromyographic (EMG) signals remains a significant challenge in assistive robotics, where traditional pattern recognition approaches struggle with continuous operation requirements. This work presents a Twin Delayed Deep Deterministic Policy Gradient (TD3) reinforcement learning framework trained to interpret EMG signals to motor commands for a 3D-printed prosthetic hand. The agent architecture and reward structure underwent iterative refinement through experimental observations to optimize proportional operation behavior. Training utilized a dataset from 12 participants provided by the Alan Turing Laboratory at Escuela Politécnica Nacional, with evaluation conducted in both simulated and hardware environments. Quantitative assessment revealed a mean absolute error between 10-15% in tracking EMG-commanded positions, with hardware testing demonstrating finger actuation capabilities and limitations that inform future development. These results establish a foundation for exploring deep reinforcement learning approaches in EMG-based prosthetic operation.

1 INTRODUCTION

In recent years, the scientific community has shown a growing interest in the application of Artificial Intelligence (AI) to the field of bionics, specifically in the development of prostheses for individuals with transradial amputations.

From a social perspective, these advancements strive to restore motor functionality and enhance the quality of life for users through accessible technological solutions.

Nevertheless, contemporary hand prostheses still face significant operation challenges. In many cases, these devices are restricted to a reduced set of discrete actions, which constrains the user's ability to precisely modulate variables such as force, velocity, or grip intensity.

These limitations are further exacerbated by the lack of adaptability in traditional operation systems. Variations in user physiology, as well as the length and specific characteristics of the amputation, hinder generalized calibration.

To address these operation limitations, surface Electromyography (sEMG) has emerged as a viable alternative for capturing motor intention. While sEMG signals enable non-invasive acquisition via skin-surface electrodes, they are characterized by high dimensionality and stochastic behavior. Given the complexity inherent to sEMG signals, Reinforcement Learning (RL) has been explored as a strategy to address continuous operation problems derived from complex data.

Specifically, Policy Gradient-based approaches have shown potential for learning policies directly from sEMG signals. Algorithms such as Twin Delayed Deep Deterministic Policy Gradient (TD3) have been proposed to mitigate training instability.

The primary contribution of this work lies in exploring the application of a Twin Delayed Deep Deterministic Policy Gradient (TD3) framework for EMG-based prosthetic operation through evaluation in both simulated and physical environments.

Motor precision is quantified using Mean Absolute Error (MAE) between commanded and actual finger positions across four independent motor channels, providing a measurable assessment of the agent's tracking capabilities.

To achieve this objective, the study follows a fourstage methodology: literature review, agent training using pre-recorded sEMG data, simulation evaluation, and preliminary validation on a 3D-printed prosthesis.

The remainder of this paper follows this methodological progression, beginning with the theoretical framework, followed by technical implementation, experimental results, and concluding with final remarks and future research recommendations.

2 RELATED WORK

The development of upper limb prostheses has evolved from purely mechanical designs to sophisticated AI-integrated systems. This section examines this progression, identifying key technological milestones and persistent control challenges that motivate the present work.

2.1 Developments in Mechanical and Myoelectric Control

Early prosthetic innovations prioritized mechanical simplicity and cost reduction. Ten Kate et al. demonstrate in (Ten Kate et al., 2017) that 3D-printed prostheses could be manufactured for approximately $500, dramatically improving accessibility compared to conventional devices.

While this democratization represented a significant advance, these mechanically-driven designs struggled to achieve the fluid, adaptive movements necessary for complex manipulation tasks. To address these limitations, researchers explored myoelectric control systems.

Guacho Rivera reports in (Rivera, 2022) a 95% gesture recognition accuracy using pre-programmed EMG patterns, with 2ms sampling rates and sub-second execution latency. However, their approach required lengthy user-specific calibration and remained constrained to discrete gestures.

Concurrent work emphasized mechanical sophistication. As described in (Vargas et al., 2020), Vargas et al. developed a five-degree-of-freedom transradial prosthesis that could support up to 4kg with minimal noise output (¡55¡ 55 ¡55 dB). While aiming for natural mobility, their results highlight a critical balance in response time.

The authors identified execution delays on the order of one second may reduce usability and task success in grasping scenarios.

These approaches, while functional and cost-effective, exposed a critical limitation: the inability to continuously modulate grasp force, velocity, and aperture. This capability is essential for naturalistic prosthetic control. This gap motivated the exploration of adaptive learning methods, as discussed in the following section.

2.2 Reinforcement Learning in Prosthetic Control

The integration of reinforcement learning into anthropomorphic robotic systems emerged as a promising solution to overcome the limitations of pre-programmed control schemes.

Peters and Schaal establish in (Peters and Schaal, 2008) foundational work in applying policy gradient methods to motor skill learning, demonstrating that RL could scale to high dimensional continuous state action spaces using motor primitives and natural actor critic algorithms.

Their Episodic Natural Actor-Critic algorithm outperformed previous methods by at least an order of magnitude. It proved highly effective in learning complex motor skills with anthropomorphic robot arms.

Garikayi et al. identify in (Garikayi et al., 2016) key challenges in translating successful in-silico RL research to clinical applications. They noted that many proposed solutions remained simulation-based with limited hardware prototypes, reducing their practical applicability.

Specific challenges in real-world prosthetic RL applications are further characterized by Salwan et al. in (Salwan et al., 2020). They highlighted that model-free algorithms like PPO and DDPG become inefficient and slow in higher-dimensional environments typical of prosthetic control.

Their comparative analysis found that a mix of model-based and model-free approaches performed best despite these challenges. Despite these efficiency concerns, notable success in upper limb control was achieved by Gao et al. in (Gao et al., 2020).

They implemented Deep Deterministic Policy Gradient (DDPG) combined with Hindsight Experience Replay (HER). Their framework for continuous shared control demonstrated that RL can effectively bridge the gap between human intention and robotic execution.

By assigning high-level planning to the user and precise finger movements to the RL agent, DDPG handled high-dimensional spaces. They managed a 20 degree of freedom system to achieve stable grasping with a 90.7% success rate.

A comparative analysis using an EMG dataset from 12 participants is conducted by Araque et al. in (Araque et al., 2024). They contrasted Deep

Q-Learning (DQN) with traditional Reinforcement Learning (RL) approaches for myoelectric prosthesis control.

The DQN agent achieved a 78% success rate compared to 86% for the RL agent. This lower performance was attributed to high sensitivity to hyperparameter tuning and the complexity of mapping noisy EMG signals to precise motor actions.

Despite these limitations, the authors highlighted the potential of DQN for long term adaptability. They recommended research into hybrid models, real-time continuous calibration, and more realistic simulations that reflect real world conditions.

Hutabarat et al. propose in (Hutabarat et al., 2020) a reward shaping approach for reinforcement Q-learning control in a semi-active prosthetic knee, designing the reward function as a function of subject-specific knee angle trajectory.

Their results indicated that a multi-component reward function outperformed conventional single reward formulations in terms of convergence speed and trajectory tracking accuracy, establishing reward shaping as a relevant design consideration for RL-based prosthetic control.

The combination of supervised learning and reinforcement learning for myoelectric control is explored by Freitag et al. in (Freitag et al., 2024), using a pretrained classifier fine-tuned with dynamic EMG data obtained during real-time interaction.

Their approach reported improvements in decoding accuracy during usage-based evaluation, suggesting that incorporating interaction data into the training process may contribute to more robust intent interpretation in bionic limb applications.

2.3 Twin-Delayed Deep Deterministic Policy Gradient for Continuous Control

The TD3 algorithm, introduced by Fujimoto et al. in (Fujimoto et al., 2018), addresses critical limitations in actor-critic methods. It mitigates function approximation errors that lead to overestimated value estimates.

By employing Clipped Double Q-learning with paired critics and delayed policy updates, TD3 reduces per-update error. This improves stability, making it suitable for continuous control tasks in complex, high-dimensional environments.

Furthermore, through the addition of target policy smoothing, the algorithm effectively reduces variance. This allows it to outperform state-of-the-art methods across all tested OpenAI Gym environments.

Sacchi et al. demonstrate in (Sacchi et al., 2021) the effectiveness of the TD3 algorithm in achieving gait symmetry for trans-femoral amputated patients. Their research showed the algorithm can adapt to various walking velocities without requiring online parameter retuning.

This represents a significant advantage over classical model-based strategies. By employing a model-free approach, they successfully generated continuous velocity control signals for prosthetic joints in realistic CoppeliaSim simulations.

While their focus remained on lower-limb systems, this work establishes a crucial precedent for using TD3. It suggests a viable framework for extension to other complex robotic limb applications, including upper-limb systems.

Further supporting this algorithmic selection, Shen conducts in (Shen, 2024) a comparative analysis of DDPG and TD3 in continuous locomotion control tasks, finding that TD3's twin critic networks, target policy smoothing, and delayed policy updates contributed to more stable convergence and reduced overestimation bias compared to DDPG.

These characteristics informed the decision to adopt TD3 as the basis for the present work. Building on these foundations, our work extends TD3's application to upper limb prostheses. We specifically address the limitations of discrete action approaches, such as the one implemented by Araque et al. in (Araque et al., 2024).

Discrete systems can struggle with the high sensitivity of mapping noisy EMG signals to precise motor actions. Unlike these methods, our TD3-based solution enables smooth, continuous finger movements that adapt dynamically to varying grasp requirements.

This addresses the high-dimensional continuous action space challenge identified by Salwan et al. in (Salwan et al., 2020).

We leverage TD3's proven stability improvements over DDPG, such as reduced overestimation bias as demonstrated by Fujimoto et al. in (Fujimoto et al., 2018).

By combining the continuous control shown effective in lower limbs (Sacchi et al., 2021) with the precision requirements of upper limb manipulation (Gao et al., 2020), this research fills a critical gap. It connects traditional designs with the need for adaptive, natural control.

3 PROPOSED APPROACH

This section describes the computational architecture, experimental framework, and iterative refinement process employed to develop a TD3-based operation system for upper limb prostheses. The methodology encompasses the dataset characteristics, observation and action space definitions, neural network architecture, hyperparameter configuration, and the evolution of the reward function design.

3.1 Dataset and Computational Framework

This research utilized the Denis Dataset (Araque et al., 2024), comprising EMG recordings from 12 participants with 100 samples per movement type, totaling 7,200 samples. Each recording captured 5 seconds of natural hand movements (opening, closing, pinching) using eight channels from a Myo armband positioned on the forearm.

The raw EMG signals underwent feature extraction to generate a 40-dimensional feature vector by computing five features per channel (Standard Deviation, Integral Absolute Envelope, Mean Absolute Value, EMG Energy, and Root Mean Square) across eight channels. These EMG features were temporally synchronized with the flexion glove data, which was reduced from nine individual flex sensors to four reference values by summing sensors corresponding to each prosthetic finger.

However, the four reference values from the flexion glove were not provided to the agent's observation space and served exclusively as the reward target signal, ensuring that the agent's exploration and learning relied solely on myoelectric interpretation rather than access to direct ground-truth flexion labels.

3.2 Environmental Interface: Observation and Action Spaces

The observation space constitutes the agent's sensory input, providing a 44-dimensional state vector consisting exclusively of 40 dimensional EMG feature vector representing normalized muscular intent, and 4 dimensional motor encoder positions providing kinematic feedback of the prosthesis state. The EMG features were normalized to a zero-mean distribution with unit standard deviation, producing values primarily in the range $[-3,+3]$. Motor encoder readings were scaled by device-specific factors $[26500, 11500, 8500, 9000]$ to normalize positions into the range $[-1,1]$, enabling the agent to perceive both neurological intent and physical reality simultaneously.

The action space, detailed in Table 2, defines continuous velocity commands for four DC motors controlled via Pulse Width Modulation (PWM).

Table 1: Observation Space Components.

Component	Dim	Range
EMG Features	40	$[-3,3]$
Motor Encoders	4	$[-1,1]$
Total State	44	—

Each action dimension corresponds to a motor's velocity change, bounded within $[-1,+1]$ and subsequently scaled to PWM magnitude $[-255,+255]$ by element-wise multiplication with the speed vector $[255, 255, 255, 255]$. This continuous velocity based control enables proportional motor operation.

Table 2: Action Space Interface.

Component	Dim	Range
Motor Actions	4	$[-1,+1]$
PWM Output	4	$[-255,+255]$

3.3 TD3 Agent Architecture and Configuration

The agent-environment configuration followed the foundational framework for reinforcement learning systems established by (Sutton and Barto, 2018), defining the observation space, action space, and reward structure as the primary interface between the agent and the prosthetic hardware.

The TD3 agent implementation consisted of an actor network and twin critic networks, designed to mitigate overestimation bias inherent in actor-critic methods (Fujimoto et al., 2018). The agent architecture and training configuration underwent multiple iterative refinements throughout the experimental phase, where each identified behavioral limitation informed subsequent design decisions.

Initial configurations employed sigmoid activation functions across the hidden layers; however, this introduced a premature convergence problem that caused abrupt learning collapse.

Drawing on the activation function analysis by (Rasamoelina et al., 2020) and architectural precedents from robotic manipulation implementations such as (Fernandez-Fernandez et al., 2022; Hsieh and Chang, 2024), the architecture was revised to incorporate ReLU activations in the hidden layers, followed by a hyperbolic tangent (tanh) output layer constraining actions to [-1,+1].

As training progressed, a secondary challenge emerged: while the agent successfully maximized cumulative rewards, it developed a non-proportional operation policy driven by reward hacking.

High incentives for goal maintenance combined with insufficient smoothness penalties encouraged brute-force strategies regardless of EMG signal intensity.

Table 3 presents the hyperparameter adjustments developed to address this behavior. Key modifications included increasing the minibatch size from 64 to 256 for more stable gradient updates, reducing the actor learning rate to 1×10^{-4} to prevent abrupt policy changes.

Additional adjustments included expanding the experience buffer to one million samples for more diverse learning experiences, and introducing exploration noise decay to transition from high exploration to precise exploitation.

The actor network comprised three fully connected layers with this refined activation scheme, while each twin critic employed independent dual-branch architectures processing state-action pairs through separate pathways before merging, with each branch containing 64 units.

This architecture enables critics to decouple high-level EMG feature extraction from low-level action evaluation, providing robust value estimates for policy optimization.

Table 3: TD3 Hyperparameters Configuration.

Hyperparameter	Value
Sample Time	0.2 s
Minibatch Size	256
Experience Buffer	1×10^{6}
Actor Learn Rate	1×10^{-4}
Critic Learn Rate	1×10^{-3}
Discount Factor	0.95
Noise Std Dev	0.4
Noise Decay Rate	1×10^{-5}
Target Update Freq	2

The reward function employed in this study underwent iterative refinement based on observed agent behavior, consistent with established reward design principles (Devidze, 2025). The final implementation, presented in Algorithm 1, incorporates multiple components designed to encourage desirable behaviors while penalizing operational deficiencies. The function evaluates direction correctness, distance minimization, movement smoothness, and goal achievement across four independent motor channels. Table 4 details the reward parameters used in the implementation.

The distance penalty factor ($k=0.8$) emphasizes positional accuracy, while the smoothness penalty ($\alpha=0.165$) penalizes abrupt action changes, promoting gradual movements.

Table 4: Reward Function Parameters.

Component	Value
Correct Action Reward	1.0
Inverse Movement Penalty	-1.0
Wrong Stop Penalty	-1.0
Movement Incentive	2.0
Distance Penalty Factor (k)	0.8
Smoothness Penalty (α)	0.165

Data: $posFlex_t$, $posFlex_{t-1}$, $flexConv_t$, a_t, a_{t-1}
Result: Scalar reward R_{total}, per-motor vector $\vec{R}$
Initialize $\vec{R} \leftarrow \mathbf{0}_{1\times4}$
Define rewards: $r_{good}=1.0$, $r_{inv}=-1.0$, $r_{wrong}=-1.0$, $r_{inc}=2.0$
Define penalties: $k=0.8$, $\alpha=0.165$
for *each motor* $i=1$ *to* 4 **do**
  if $posFlex_{t-1,i} < posFlex_{t,i}$ **then** $c_i \leftarrow 1$
  else if $posFlex_{t-1,i} > posFlex_{t,i}$ **then** $c_i \leftarrow -1$
  else $c_i \leftarrow 0$
  if $a_{t,i}=c_i$ **then**
    $R_i \leftarrow (a_{t,i}\neq 0)\ ?\ r_{good}\ :\ 0$
  else
    $R_i \leftarrow (a_{t,i}=0)\ ?\ r_{wrong}\ :\ r_{inv}$
  end
end
if $\exists i: a_{t,i}\neq 0$ **then**
  $\vec{R} \leftarrow \vec{R}+r_{inc}$
end
for *each motor* $i=1$ *to* 4 **do**
  $dist_i \leftarrow |posFlex_{t,i}-flexConv_i|$
  $smooth_i \leftarrow |a_{t,i}-a_{t-1,i}|$
  $R_i \leftarrow R_i - k\cdot dist_i - \alpha\cdot smooth_i$
end
$R_{total} \leftarrow \text{mean}(\vec{R})$
return $R_{total}, \vec{R}$

Algorithm 1: Distance Reward Function Pseudocode

3.4 Hardware-Software Communication Architecture

The operational architecture establishes a closed-loop pipeline connecting neural inference with hardware execution. Normalized EMG signals serve as input to the trained TD3 agent, which outputs Motor Control Commands (MCCs) transmitted as PWM signals through an Arduino microcontroller to DC motor encoders. This architecture enables real-time continuous operation where motor velocities adapt dynamically to EMG signal variations, as detailed in Figure 1.

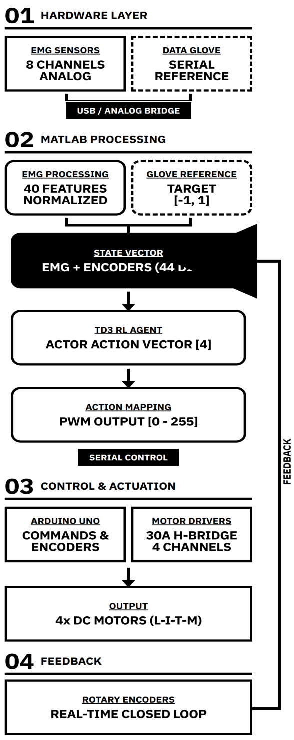

Figure 1: Operational architecture of the system.

4 RESULTS

This section presents the experimental outcomes obtained from the TD3-based prosthetic operation system, detailing training performance metrics, simulation behavior analysis, and real-time hardware validation results.

4.1 Training Performance and Convergence

Following the architectural refinements and reward function modifications described in 3.3, the agent underwent intensive training evaluation. Initial validation across 10,000 episodes showed stabilization of cumulative episodic rewards, as illustrated in Figure 2. The training trajectory exhibited reduced variance compared to earlier configurations, with the average reward converging to stable values within the first 6,000 episodes. This convergence behavior is consistent with the expected effects of the increased minibatch size (256) and reduced actor learning rate 10≏4 on policy update stability across diverse EMG signal patterns.

Based on these results, training was extended to 100,000 episodes using the complete dataset encompassing all twelve participants. Figure 3 shows sustained convergence despite the added complexity of multi-subject data. The larger minibatch sizes and optimized learning rates contributed to policy stability across the participant population. Notably, the Avergare Reward (dark blue line) tracked the cumulative reward progression, suggesting value function learning alongside policy optimization. The sustained upward trend and reduced variance indicate that the agent was exposed to sufficiently varied EMG patterns to begin capturing relationships between muscular activation and proportional motor output, though generalization beyond the training set was not formally evaluated.

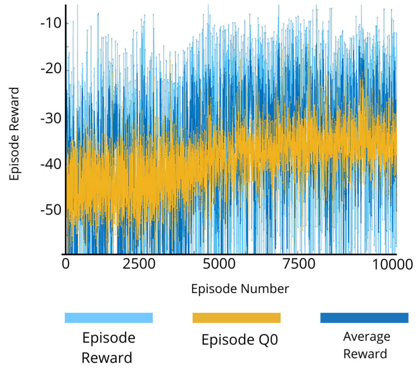

Figure 2: TD3 Training Curves (10k episodes): Orange line= Critic Q-value at initial state (Episode Q0); dark blue line = mean episodic reward; cyan line = per-step rewards.

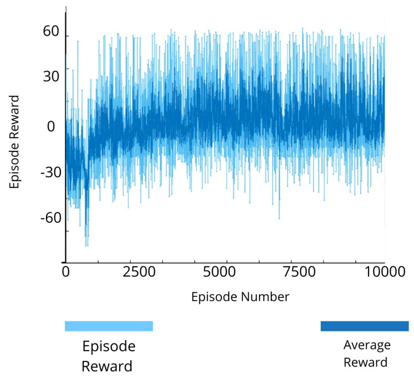

Figure 3: TD3 Training Curves (100k episodes): Dark blue line = mean episodic reward; Cyan line = per-step rewards..

4.2 Simulation Results: Proportional Operation Achievement

Trajectory-tracking simulations provided qualitative evidence of improved operational behavior following the architectural modifications.

Each step corresponds to a single control cycle of

0.2 seconds, so the maximum episode duration of 50 steps represents 10 seconds of continuous operation; step 16, for instance, corresponds to 3.2 seconds of execution.

At each step, the agent receives 40 EMG features extracted from the simulated signal, issues a normalized motor command, and receives encoder feedback before the reward is computed. The horizontal axis therefore reflects elapsed time in discrete 0.2-second intervals, as defined by the sampling period in the environment configuration.

Figures 4 and 5 illustrate the correlation between the desired reference trajectory and the actual prosthesis position during hand extension and flexion tasks respectively.

The vertical axis represents normalized position in the range $[-1, 1]$, where the orange trace corresponds to the EMG-simulated reference and the blue trace corresponds to the encoder-measured prosthesis position.

Each subplot corresponds to an independent motor channel, and the red markers overlaid on the blue trace indicate the discrete actions selected by the TD3 agent at each decision step.

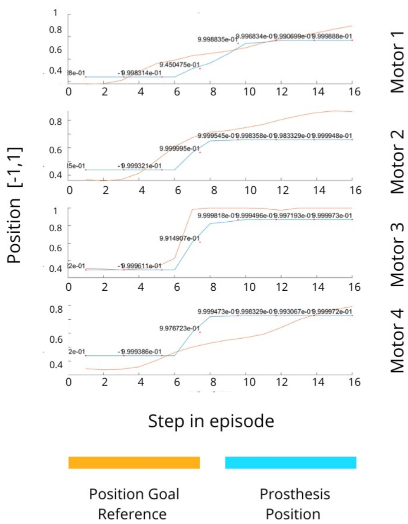

Figure 4: EMG-motor correlation during hand extension. Motor actions track EMG intensity variations.

The motor responses exhibited greater proportional modulation compared to the aggressive, all-or-nothing patterns observed in early training phases. Notably, the refined architecture allowed the agent to produce responses to hand opening (extension) commands, a behavior that had been largely absent in initial configurations where the reward structure inadvertently discouraged extension movements in favor of rapid closure strategies.

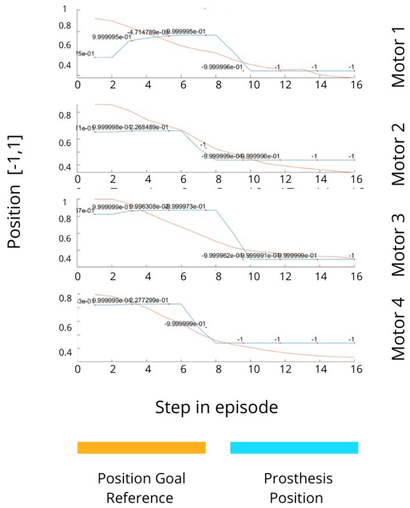

Figure 5: EMG-motor correlation during hand flexion showing more continuous operation.

To quantitatively assess tracking performance in simulation, Mean Absolute Error (MAE) was computed as the average positional difference between the reference glove trajectory and the simulated prosthesis position across all motor channels. Figure 6 shows the MAE evolution across 48 evaluation episodes. The agent achieved a mean MAE of 0.1553 (normalized scale), with minimum and maximum values of 0.0748 and 0.2676 respectively, indicating approximately 15% average positional error. The observed variability between episodes reflects the agent's adaptation to different EMG signal patterns present in the dataset.

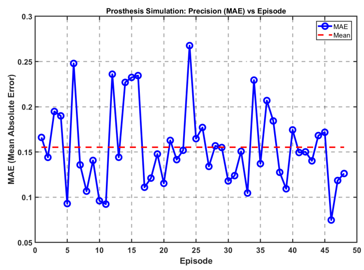

Figure 6: Tracking precision across 48 evaluation episodes showing MAE between reference and simulated trajectories.

4.3 Real-Time Hardware Validation

The TD3-based operation system underwent preliminary technical validation with live subjects to assess real-time operational capabilities using hardware provided by the Alan Turing Artificial Intelligence and Computer Vision Research Laboratory. Different outcomes emerged due to individual physical characteristics and electronic noise present in biological signals.

Under these hardware conditions, the agent produced responses to primary motor intentions in several cases. Figures 7 and 8 show real-time trajectory tracking during hand extension (opening) and flexion (closing) phases respectively, indicating that the trained model was able to drive physical actuators in real-time, though with reduced consistency compared to simulation conditions.

Figure 7: Real-time trajectory tracking during hand extension (opening gesture).

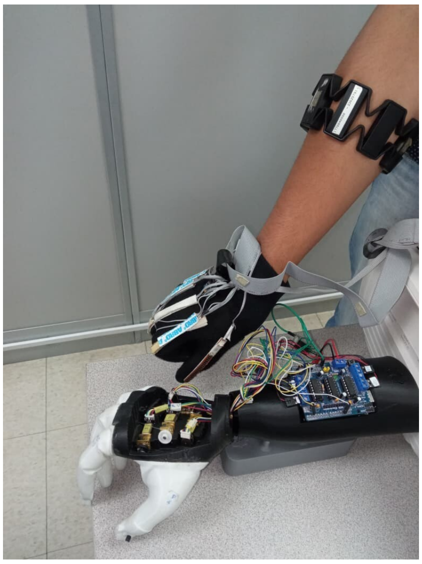

Figure 8: Real-time trajectory tracking during hand flexion (closing gesture).

Finger-specific performance varied considerably during hardware validation. These results must be interpreted in the context of sensor displacement and noise issues, which heavily influenced the observed behavior of each finger. Only the middle finger exhibited a reliable response, following stimuli with reasonable fidelity under the most stable signal conditions of the test.

The thumb showed acceptable responsiveness, though its performance was partially affected by noise at the base of the wrist.

The remaining fingers showed inconsistent or limited results. The index finger exhibited restricted and dampened movement throughout the validation.

The ring and pinky fingers behaved erratically, exhibiting aggressive yet inconsistent actuation during extension and struggling most with hand-opening commands.

These results suggest that, under the current hardware conditions, reliable independent operation of these fingers was not achieved.

The source code, dataset, and supplementary resources developed in this work are publicly available at the following link, located in the matlab code folder. https://github.com/laboratorioAI/EMG_Prosthesis_TD3

5 DISCUSSION

This section examines the key findings from the TD3-based prosthetic operation system, analyzing both technical achievements and practical limitations encountered during development and validation. The discussion addresses training dynamics and reward function sensitivity, highlights the system's contributions to proportional EMG-based operation, evaluates limitations exposed during hardware testing, and explores theoretical and practical implications for future development of reinforcement learning approaches in myoelectric prosthetic systems.

5.1 Training Evolution and Behavioral Improvement

Reward function design proved to be one of the most critical factors throughout the development process. Early training phases revealed that quantitative reward maximization does not necessarily correlate with qualitative operational suitability, as the agent found mathematically optimal but functionally inappropriate strategies by exploiting imbalanced reward weights. Adjusting the distance penalty weight from

0.05 to 6.0, combined with the transition to ReLU activation functions, shifted the optimization landscape toward continuous error minimization rather than discrete goal achievement. These modifications reduced the tendency toward abrupt, all-or-nothing actuation and produced more graded motor responses over the course of training. The expanded minibatch size (256) contributed to more stable policy updates by exposing the agent to a wider variety of state-action transitions before each update, which appeared to reduce premature convergence to binary strategies. The continued improvement observed throughout 100,000 episodes suggests that convergence had not yet saturated, indicating that extended training remains a practical avenue for refinement.

5.2 Quantitative Tracking Performance

The mean MAE of 0.1553 across 48 evaluation episodes corresponds to approximately 15% average positional error relative to reference trajectories in simulation.

The minimum observed MAE of 0.0748 suggests that tighter tracking is possible under certain signal conditions, while the maximum MAE of 0.2676 indicates notable variability across episodes. This performance range reflects sensitivity to input signal characteristics and movement patterns present in the dataset.

The observed variability may stem from the inherent challenge of interpreting stochastic EMG signals, where signal quality, participant-specific activation patterns, and movement dynamics influence tracking consistency.

The MAE metric provides a quantitative reference for simulation performance and highlights areas where further refinement may be beneficial, particularly regarding signal variability handling and consistency across diverse EMG patterns.

5.3 System Strengths and Contributions

The primary technical contribution of this work is achieving trajectory-tracking behavior that reacts to EMG signal intensity in a graded manner, in contrast to the discrete, pre-programmed control schemes typically employed in traditional myoelectric prostheses. The policy was trained on data from twelve participants with varying EMG characteristics without participant-specific fine-tuning. While formal generalization tests were not conducted, the cross-subject training setup suggests the policy was exposed to a degree of signal variability across participants, though the extent to which this translates to generalization was not formally evaluated. The iterative development process also provides methodological insight into reward function design for human-machine interfaces, particularly regarding the balance between competing optimization objectives when user comfort is a relevant consideration.

5.4 Limitations and Threats to Validity

Real-time hardware validation exposed challenges that limit the interpretation of the simulation results. EMG signal noise represents a significant obstacle in uncontrolled environments, where external interference, electrode contact variability, and biological signal instability affect control consistency in ways not captured during training. The finger-specific performance disparities indicate that the obtained movement, while capturing the general direction of intended actions, did not achieve consistent actuation across all digits. Sensor placement variations and hardware limitations contributed to erratic behavior particularly during extension movements. Trajectory tracking, while improved compared to binary control, did not reach smooth or fully coordinated movement. The agent's output remains sensitive to input signal quality, suggesting that additional preprocessing or adaptive filtering may be necessary before considering any deployment beyond controlled experimental settings.

5.5 Practical and Theoretical Implications

The sensitivity to reward function design observed throughout this investigation reinforces the need for iterative refinement guided by both quantitative metrics and qualitative behavioral assessment. Small changes in reward weights produced disproportionate shifts in emergent behavior, suggesting that this balance requires careful tuning for each specific application context. The gap between simulation performance and hardware results highlights the importance of incorporating real-world signal variability into the training environment. Preprocessing stages that reduce noise and normalize input signals could help close this gap and improve the consistency of motor responses during physical operation. Extended training regimes combined with systematic hyperparameter optimization could identify more suitable configurations for this application. Investigating reward function formulations that explicitly incorporate safety and user comfort constraints alongside trajectory tracking may also contribute to more suitable policies for practical use.

6 CONCLUSIONS

This research explored the application of TD3 reinforcement learning for continuous, proportional control in upper limb prosthetic systems driven by electromyographic signals, addressing the challenge of moving beyond binary, pre-programmed myoelectric control schemes.

6.1 Summary of Contributions

The development process indicated that reward function design and architectural choices have a visible influence on emergent agent behavior in this type of application. The transition to ReLU activation functions combined with recalibrated reward weights contributed to more graded motor responses, reducing the abrupt actuation patterns observed in earlier configurations.

Hardware validation indicated that the trained model produced responses aligned with the general direction of intended movements in several cases. However, consistent actuation across all digits was not achieved, and the results should be interpreted as a preliminary observation rather than a validation of the approach.

6.2 Practical Value and Limitations

This work constitutes an initial experimental evaluation of TD3 for EMG-driven prosthetic control in a low-cost hardware setting, rather than a definitive methodological foundation. The results suggest that reinforcement learning is a relevant direction for this type of application, though the gap between simulation behavior and hardware performance remains a primary unresolved challenge. Signal noise and electrode variability were the most disruptive factors during real-time operation, and their effect on finger-level control was considerable. Addressing these at the signal acquisition level appears to be a necessary condition for any meaningful progress beyond the current results.

6.3 Future Directions

Extended training beyond 100,000 episodes remains an unexplored variable in this work, as convergence had not saturated by the end of training. Systematic hyperparameter optimization could also identify configurations better suited to the variability present in real-world EMG signals. Incorporating real-world signal conditions into the training environment, rather than relying solely on offline datasets, may help reduce the inconsistencies observed during hardware operation. Reward function formulations that explicitly account for hardware constraints and signal noise could further contribute to more stable policies under physical deployment conditions.

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APPENDIX

If any, the appendix should appear directly after the references without numbering, and not on a new page. To do so please use the following command: *section**{*APPENDIX*}